Intraoperative Opioid Choice in Adult and Paediatric Tonsillectomy: A Retrospective Electronic Patient Record Cohort Study

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Abstract

Background

Severe pain following tonsillectomy is common and an important cause of delays to discharge and secondary postoperative complications. Clinical guidelines offer a range of perioperative analgesia recommendations but the optimal intraoperative opioid strategy remains unknown. This study aimed to evaluate existing variation in perioperative analgesic management and its impact on postoperative outcomes at a high-volume surgical centre.

Methods

A retrospective cohort study was performed to identify all patients undergoing tonsillectomy, adenoidectomy or combined adenotonsillectomy over a 7-year period using routinely collected data from the electronic patient record. Medication exposure and outcomes were extracted, and patients grouped on analgesic exposure to simple analgesia (paracetamol, dexamethasone and diclofenac), plus or minus additional morphine, fentanyl or both. Descriptive analysis of the analgesic exposure, treatment groups and multilevel modelling was performed.

Results

3,656 patients were included, of which 2,259 (61.8%) were paediatric patients. 82.9% of adults received all three components of simple analgesia compared to 72.0% of paediatric patients. Fentanyl was the most common intraoperative opioid in the paediatric cohort (95.4% of cases) but adults received a mix of strategies, with 49.0% receiving fentanyl only (n = 685), 22.8% morphine only (n = 318), 17.7% receiving both (n = 247), and 10.5% receiving neither (n = 147). There was no significant difference in the need for recovery opioids between treatment groups for adult patients, however simple analgesia with both morphine and fentanyl resulted in a reduced time in recovery (-42.1 minutes, 95% CI -66.7 to -17.5).

Conclusions

Intraoperative opioid strategies varied for adults undergoing tonsillectomy procedures, but were more homogeneous in paediatric patients. When accounting for patient, surgical and anaesthetic factors, a combined opioid regime of fentanyl and morphine with complete simple analgesia was associated with better odds of same day discharge and shorter time in hospital. This treatment plan was not associated with an increased odds of opioid prescriptions on discharge nor representations to the emergency department. Further work to address the limitations of data quality and possible causality is needed to build evidence for intra-operative opioids in tonsillectomy.

Keywords:

Tonsillectomy, Analgesia, Pain, Opioids, Retrospective, Multi-level modelling

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1.1 Background

Tonsillectomy and adenoidectomy are high-volume surgical procedures. In the UK nearly 50,000 tonsillectomies are performed each year, typically for managing recurrent tonsillitis and/or sleep disordered breathing.1,2 Procedures may be undertaken in combination (adenotonsillectomy) or isolation, depending on the indication, and most are day case procedures.

1.2 Methods

1.2.1 Design, Setting and Population

A single centre, retrospective electronic patient record (EPR) cohort study. Patients who underwent elective tonsillectomy, adenoidectomy, or combined adenotonsillectomy were included. Procedures were identified using relevant OPCS Classification of Interventions and Procedures codes (**?@sec-sm-data-spec**).

1.3 Results

The results have shown one result Figure 1.1.

1.4 List os things

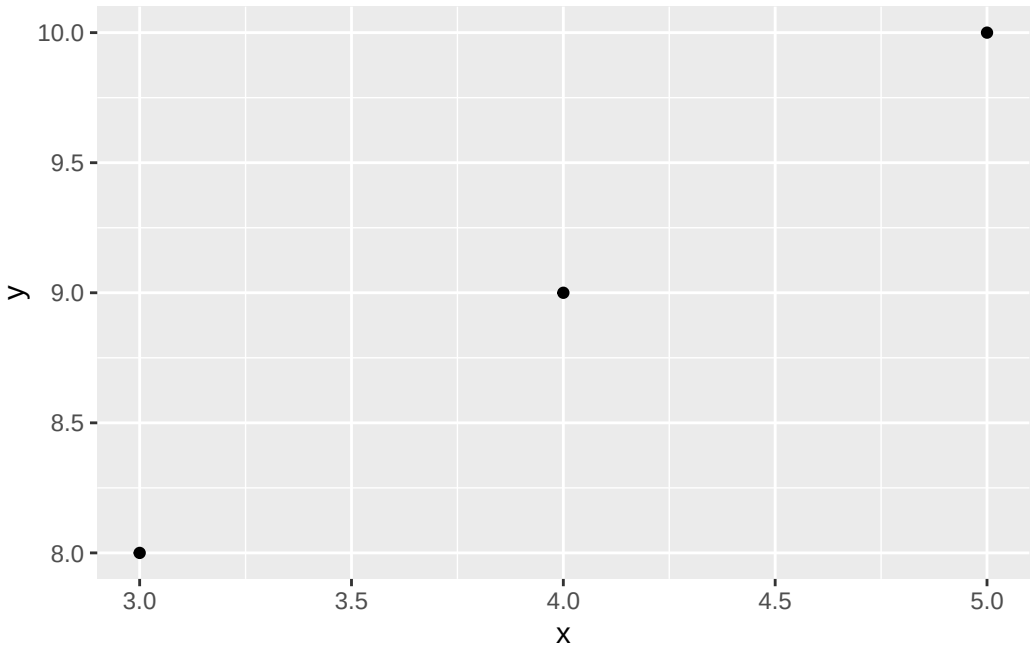

Figure 1.1: The example plot that shows three dots

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